

Literature Answer to the Continuous-Parameter Besov Katznelson–Tzafriri Question from arXiv:2204.13411

FA/Banach run packet

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Status

This is a literature-already-answered packet. It records that the continuous-parameter Besov-calculus question noted in Batty–Seifert [1] is answered by the companion/later paper Batty–Seifert [2]. It is not an original proof from this run.

Original Source Signal

In Section 5.1 of [1], after discussing the analytic Besov algebra $\mathcal{B}(\mathbb{C}_+)$ and the corresponding calculus for negative generators of bounded C_0 -semigroups, the authors write that the results in the discrete Besov section are analogous to the semigroup Besov-calculus results except for Theorem 2.7. They then state that one can formulate a continuous-parameter version of that theorem, but that they had not determined whether it is true.

The discrete theorem in question says that if T is power-bounded, has a bounded $\mathcal{B}(\mathbb{D})$ -calculus, and $f \in \mathcal{B}(\mathbb{D})$ vanishes on $\sigma(T) \cap \mathbb{T}$, then

$$\lim_{n \rightarrow \infty} \|T^n f(T)\| = 0.$$

Supporting Answer

The paper [2] has exactly the title *A continuous-parameter Katznelson–Tzafriri theorem for algebras of analytic functions*. Its abstract states that it proves a continuous-parameter version of the recent Besov-calculus Katznelson–Tzafriri theorem for power-bounded operators. Its introduction cites the survey [1] and says that the authors subsequently proved the corresponding continuous-parameter result.

The principal Besov statement is Theorem 1.2 of [2]:

Literature Answer. *Let $-A$ be the generator of a bounded C_0 -semigroup $(T(t))_{t \geq 0}$ on a Banach space and suppose that A has a bounded $\mathcal{B}(\mathbb{C}_+)$ -calculus. If $f \in \mathcal{B}(\mathbb{C}_+)$ vanishes on $\sigma(A) \cap i\mathbb{R}$ and*

$$\lim_{\substack{z \in \mathbb{C}_+ \\ |z| \rightarrow \infty}} f(z) = 0,$$

then

$$\lim_{t \rightarrow \infty} \|T(t)f(A)\| = 0.$$

Theorem 2.1 of [2] gives a larger-algebra version: if $\mathcal{A}$ is a Banach algebra with $\mathcal{B}(\mathbb{C}_+) \subseteq \mathcal{A} \subseteq H^\infty(\mathbb{C}_+)$, $-A$ generates a bounded C_0 -semigroup and has a bounded $\mathcal{A}$ -calculus, and f lies in the closure of Laplace transforms in $\mathcal{A}$ while vanishing on $\sigma(A) \cap i\mathbb{R}$, then the same operator-norm decay follows.

Scope and Identification

This answers the exact unresolved continuous-parameter analogue signalled in [1]. The vanishing-at-infinity hypothesis in Theorem 1.2 is part of the correct continuous-parameter formulation for unbounded generators; the supporting paper explains that without it constant functions would give counterexamples when the semigroup is bounded but not exponentially stable.

The answer source is separate from the survey source. It also explicitly knows the surrounding problem context: it cites the survey as [1], cites the discrete Besov theorem as [3], and states that it proves the corresponding continuous-parameter theorem.

Verification Notes

Local source inspection used:

- `data/parsed/arxiv_sources/2204.13411/source.tex`, lines 701–704, for the source signal.
- `data/parsed/arxiv_sources/2204.13416/source.tex`, lines 129–130 and 155–166, for the abstract/intro match and Theorem 1.2.
- `data/parsed/arxiv_sources/2204.13416/source.tex`, lines 210–214 and 335–341, for the larger-algebra theorem and its proof conclusion.

Web search for the exact supporting title also returned arXiv:2204.13416 with the same abstract. No local registry row for this exact target existed before this packet.

References

- [1] C. Batty and D. Seifert, *Some developments around the Katznelson–Tzafriri theorem*, arXiv:2204.13411; Acta Sci. Math. (Szeged) 88 (2022), 53–84.
- [2] C. Batty and D. Seifert, *A continuous-parameter Katznelson–Tzafriri theorem for algebras of analytic functions*, arXiv:2204.13416; Studia Math. 270 (2023), 229–239.
- [3] C. Batty and D. Seifert, *A Katznelson–Tzafriri theorem for analytic Besov functions of operators*, arXiv:2201.12076.